DSP · L10

IIR design: bilinear transform and Butterworth

Cheat sheet · Day 19 · Wed 11 Nov · pack D11

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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IIR DESIGN BY BILINEAR TRANSFORM (L10)  -- take T = 1 in designs (T cancels)
Map:      s = (2/T)(1 - z^-1)/(1 + z^-1)        s=0 -> z=1,  s=inf -> z=-1
Warping:  Omega = (2/T) tan(w/2)    <->   w = 2 arctan(Omega T / 2)
          w = 2 pi f / Fs   (digital);   Omega = 2 pi f (pure analogue only)
6 STEPS: 1 digital specs wp, ws, dp, ds   2 PREWARP wp, ws -> Wp, Ws
         3 analogue |H(W)|   4 H(s)   5 bilinear -> H(z)   6 CHECK specs
BUTTERWORTH  |H(W)|^2 = 1/(1+(W/Wc)^2N)      Wc = 3-dB point = pole radius
 dp = 10^(-Ap/20) (MIN passband gain)   ds = 10^(-As/20)
 N = log[(10^(As/10)-1)/(10^(Ap/10)-1)] / (2 log(Ws/Wp))     ROUND UP
 Wc (passband exact, slide 17) = Wp / (10^(Ap/10)-1)^(1/2N)
 Wc (stopband exact, T9 Q5)    = Ws / (10^(As/10)-1)^(1/2N)   -> say which!
 LHP poles s_k = Wc exp(j(pi/2 + pi(2k+1)/(2N))), k=0..N-1;  H(s)=Wc^N/prod(s-s_k)
 N=2: s^2+1.4142Wc s+Wc^2      N=3: (s+Wc)(s^2+Wc s+Wc^2)
 N=4: (s^2+0.7654Wc s+Wc^2)(s^2+1.8478Wc s+Wc^2)
 Pair at angle th: s^2 - 2Wc cos(th) s + Wc^2
FACTOR METHOD (K = 2/T):
 s+a      -> [(K+a) + (a-K) z^-1] / (1+z^-1)
 s^2+bs+c -> [(K^2+bK+c) + 2(c-K^2) z^-1 + (K^2-bK+c) z^-2] / (1+z^-1)^2
 numerator Wc^N (1+z^-1)^N ; divide by den constant so a0 = 1
 diff eq: y(n) = -a1 y(n-1) - a2 y(n-2) ... + b0 x(n) + b1 x(n-1) ...
CHECK: gain_dB = -10 log10[1+(W/Wc)^2N] at Wp and Ws (= digital gain at wp, ws)
ORDER FROM POLE RADIUS R: Wc = R; N = log(10^(A/10)-1)/(2 log(W1/Wc)), W1 = 2 pi f
IMPULSE INVARIANCE (low priority): h(n)=T ha(nT), pole p -> e^(pT); aliases
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